Differential equations — 4.1

$y' = ky$ for y some function of t.

A differential equation is an equation involving an unknown function and one or more of its derivatives. A solution to a differential equation is this unknown function.

Example: $y' = ky$ has solutions of the form $y(t) = y_0 e^{kt}$.

Such a solution is called a general solution, whereas, $3e^{kt}$ is a particular solution.

Example: Any problem of the form

$y'(t) = f(t)$ is a differential equation.

A solution is given by an antiderivative of f. If F is such an antiderivative, then $F(t) + C$ is the general solution.

$y' = 2x$, then $y(x) = x^2 + C$ is the general solution.

$y(x) = x^2 - 1$ is a particular solution.

Example: Verify that $y(x) = e^{-3x} + 2x + 3$ is a solution of $y' + 3y = 6x + 11$.

general solution: $Ce^{-3x} + 2x + 3$

$\longrightarrow \; y'(x) = -3e^{-3x} + 2$

$y'(x) + 3y(x) = -3e^{-3x} + 2 + 3e^{-3x} + 6x + 9 = 6x + 11.$

This and previous examples are all instances of "first-order" equations in that the highest derivative involved is the first derivative.

E.g. $y''' - \sin(y') + 10y = \ln(t)$ is third-order.

Example: Find a particular solution to

$y' + 3y = 6x + 11$ passing through the point $(3, 6)$.

looking to find C such that

$Ce^{-3x} + 2x + 3$ is 6 when we plug in $x = 3$.

$$Ce^{-9} + 6 + 3 = 6 \iff Ce^{-9} = -3 \iff C = -3e^{9}$$

So the particular solution we want

is $-3e^{9} \cdot e^{-3x} + 2x + 3 = -3e^{9-3x} + 2x + 3$.

An <u>initial value problem</u> is a differential equation with a required set of initial data.

For first-order equations, this means a required value of $y(0)$.

<u>Example</u>: Solve the IVP $y' + 3y = 2x + 3$ and $y(0) = 6$.

$$\rightarrow \quad 6 = Ce^{0} + 0 + 3 \quad \Rightarrow \quad C = 3$$

$$y(t) = 3e^{-3x} + 2x + 3.$$

Example: Verify that $2e^{-2t} + e^{t}$ is the solution of the IVP $y' + 2y = 3e^{t}$, $y(0) = 3$.

$\rightarrow \quad 2e^{0} + e^{0} = 2 + 1 = 3 \checkmark$

$y'(t) = -4e^{-2t} + e^{t}$

$y' + 2y = \cancel{-4e^{-2t}} + e^{t} + \cancel{4e^{-2t}} + 2e^{t} = 3e^{t} \checkmark$